

Yashpreet

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Summary

CS graduate from VIT Bhopal (CGPA 8.47/10) specializing in autonomous AI agent systems, native desktop engineering, and production full-stack software. Built **Sanctum**, a local-first AI agent platform with 5-tier LangGraph orchestration, hybrid vector + BM25 retrieval, and MCP tool integration; and **AetherFlow**, an ultra-lightweight Rust/Tauri 2 Windows desktop engine with Win32 HWND injection and procedural Canvas 2D rendering at 60 FPS. Seeking **AI/ML Engineer**, SDE, or Systems Engineering roles to build and ship high-impact software from day one.

Education

B.Tech in Computer Science and Engineering

VIT Bhopal University, Bhopal

2026

CGPA: 8.47 / 10

Technical Skills

Languages : Python, Rust, TypeScript, JavaScript, C++, Java, SQL

AI / ML : LangGraph, LangChain, Model Context Protocol (MCP), Hybrid RAG, Ollama

Web & Frontend : React 19, Next.js, Tailwind CSS, Zustand, Canvas 2D, Web Audio API, WebSockets

Systems & Backend : Tauri 2, Win32 API, FastAPI, Node.js, Express.js, REST APIs, JWT

Data & DevOps : Qdrant (Vector DB), PostgreSQL, SQLite, MongoDB, MySQL, Docker, Git, GitHub Actions

Projects

Sanctum – Local-First Autonomous AI Agent Platform [\[Code\]](#)

Python • LangGraph • FastAPI • WebSockets • Qdrant • Ollama • MCP

- Engineered a **local-first autonomous AI agent platform** powered by a **5-tier cyclical LangGraph state machine** with SQLite checkpointing, autonomous self-correction loops, and a **74-scenario verification suite** achieving 100% pass rate.
- Designed a **hybrid retrieval engine** combining dense vector search (**Qdrant**, `all-MiniLM-L6-v2`) and sparse keyword search (**BM25**) with exponential recency decay scoring to rank and retrieve personal notes and files.
- Built an **Anthropic Model Context Protocol (MCP) client** connecting external tool servers (Filesystem, SQLite) via JSON-RPC 2.0 over standard I/O pipes, dynamically registering schemas for agent execution.
- Implemented real-time **WebSocket event streaming** (`astream_events`) with token-budgeted context summarization, alongside a **quarantine security gate** with canary token leak detection and Human-in-the-Loop approval barriers.

AetherFlow – High-Performance Desktop Engine & Live Visuals Platform [\[Code\]](#)

Rust • Tauri 2 • Win32 API • React 19 • Canvas 2D • Web Audio API • Zustand

- Engineered an ultra-lightweight Windows desktop engine in **Rust (Tauri 2)** with an **~8MB installer footprint**, utilizing native OS WebView2 to avoid the memory overhead of bundled Chromium/Electron runtimes.
- Implemented native **Win32 API desktop integration**: leveraged Windows 11 Shell messaging (`Progman 0x052C`) and `WorkerW` HWND parenting to embed render surfaces seamlessly behind active desktop icons.
- Achieved pixel-perfect multi-monitor DPI awareness via Win32 GDI and DWM APIs, enabling seamless borderless rendering across mixed-resolution displays without visual artifacts.
- Developed GPU-free **Canvas 2D procedural rendering engines** (60 FPS particle networks with mouse repulsion physics, matrix rain, synthwave grids) and real-time audio spectrum visualization via the **Web Audio API**.

Achievements & Certifications

- Earned **AWS Cloud Digital Badges** across core cloud services (compute, storage, networking, serverless).
- Certified in [Computer Networking](#) (Google) and [MERN Full Stack](#) (Ethnus).